MIL-C-788F(NU)
20 August 1981
SUPERSEDING
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25 April 1974

MILITARY SPECIFICATION

CLOTH, BRATTICE, COTTON, FLAME-RETARDANT

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

*1.1 Scope This specification covers the requirements for one type of cotton brattice cloth with a flame-retardant finish used in conjunction with an adhesive for repairing and covering the surface of fibrous glass insulation board.

2. APPLICABLE DOCUMENTS

*2.1 Issue of documents The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-P-1135 - Packaging and Packing of Duck Fabrics (Cotton; Synthetic Fiber; Cotton-Synthetic Fiber Blends)

MILITARY

MIL-A-3316 - Adhesive, Fire-Resistant, Thermal Insulation
MIL-I-742 - Insulation Board, Thermal, Fibrous Glass

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

*2.2 Other publications The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified the issue in effect on date of invitation for bids or request for proposal shall apply.

RULES AND REGULATIONS UNDER THE TEXTILE FIBER PRODUCTS IDENTIFICATION ACT

(Copies may be obtained without charge from the Federal Trade Commission, Washington DC 20580).

3. REQUIREMENTS

3.1 First article When specified, the contractor shall furnish sample unit(s) for first article inspection and approval. (see 6.2).

3.2 Materials (see 6.3)

3.2.1 Yarns The yarn shall be made of cotton of a suitable length of staple and grade which has been thoroughly cleaned, drawn, and evenly spun into singles yarn to meet the requirements of this specification.

*3.3 Physical requirements The finished cloth shall conform to the requirements shown in Table I when tested as specified in 4.6.

Table I - Physical requirements

Weight (oz/yd^2) (g per m^2)	Yarns/inch (2.5cm)		Breaking strength (lbs) (Newtons) minimum	
	Warp	Filling	Warp	Filling
13.0 (± 2.0) (441 ± 68)	24 (± 2)	20 (± 2)	120 (534)	50 (223)

3.3.1 Weave The cloth will be either a twill or plain weave.

*3.3.2 Width The width of the cloth inclusive of selvages shall be as specified (see 6.2). The width shall have a tolerance of ± 1 inch (2.5cm). When the cloth has shuttleless loom selvages, the width of the cloth shall be as specified (see 6.2) and shall be the minimum width of the cloth, excluding the selvages.

3.4 Color The color of the cloth shall be natural.

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*3.5 Finish The fabric shall be subjected to a natural finish and treated with a flame retardant that shall be compatible with the adhesive covered by Class 1 of MIL-A-3316 for drying time and adhesive strength. The adhesive shall dry to touch within a period of 4 hours and dry thoroughly to constant weight within a period of 48 hours. Specimens prepared and tested as specified in 4.6.1 shall have an average adhesive strength of not less than 3 pounds (13.4 Newtons).

*3.5.1 Flame-retardant requirements The treated cloth shall have an average time of after-flame of not more than 1.0 seconds, average after-glow time of not more than 2.0 seconds and an average length of char not more than 3.0 inches (7.6cm) when tested as specified in 4.6.

*3.6 Length and put up The minimum length of a continuous piece shall be 40 yards (36.4m). No roll shall be more than 200 yards (182m) in length. The pieces shall be put up in rolls and ticketed as specified in PPP-P-1135.

3.7 Fiber identification Each roll shall be labeled or ticketed for fiber content in accordance with the Textile Fiber Products Identification Act.

3.8 Workmanship The cloth shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.1.1 Certificate of compliance Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

*4.2 Classification of inspection The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

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*4.3 First article inspection When required, (see 6.2), the first article submitted in accordance with 3.1 shall be visually inspected for finish and appearance and shall be tested for chemical and physical properties in accordance with the applicable methods specified in 4.6.

*4.4 Quality conformance inspection Sampling for inspection shall be peformed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

*4.4.1 Component and material inspection In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

*4.4.2 Examination of the end item Defects found during examination shall be classified in accordance with 4.4.2.1 through 4.4.2.3.

*4.4.2.1 Yard-by-yard examination The required yardage of each roll shall be examined and defects clearly noticeable at normal inspection distance (3 feet) (0.9m) shall be classified as listed below. The defects found shall be counted regardless of their proximity to each other, except when two or more defects represent a single local condition of the cloth, only one shall be scored. A continuous defect shall be counted as one defect for each warp-wise (lengthwise) yard (0.9m) or fraction thereof in which it occurs. The sample unit shall be 1 linear yard (0.9m) full width of the finished cloth. The acceptable quality level shall be 4.0 defects per 100 units (yards) (91.4m). The sample size shall be in accordance with inspection level II of MIL-STD-105. The lot size shall be expressed in units of one linear yard (0.9m). The number of rolls selected shall be in accordance with Table II.

Defects

Hole, cut, tear or open place.
Smash, multiple floats, jerked in filling
Slub or knot 1/
Spot or stain
Crease or wrinkle

1/ To be scored only when it exceeds three times the normal yarn size.

*4.4.2.2 Overall examination Each defect shall be counted not more than once in each roll examined. The sample unit shall be one roll. The size (number of rolls selected as sample) and the acceptance number shall be as shown in Table II.

Defects

Baggy, ridgy, or wavy cloth
Width more or less than specified 1/
Not labeled in accordance with the
Textile Fiber Products Identification Act

1/ Shall not be scored unless it occurs for more than 5 yards (4.6m) per roll.

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Table II

Lot size (yards)	Sample size (rolls)	Acceptance number
Up to 1300 1/	3	0
1301 up to and including 3200	5	0
3201 up to and including 8,000	7	0
8,001 up to and including 22,000	10	0
22,001 up to and including 110,000	15	0
110,001 and over	25	1

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.

4.4.3 Examination for length

*4.4.3.1 Individual rolls During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards (1.8m) less than the length marked on the ticket, shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.4.3.2 Total yardage in sample The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

*4.5 Examination of packaging An examination shall be made in accordance with the provisions of PPP-P-1135 to determine whether packaging, packing and marking comply with Section 5 requirements.

*4.6 Testing of the end item The methods of testing specified in FED-STD-191 wherever applicable and as listed in Table III shall be followed. The physical and chemical values specified in Section 3 shall apply to the results of the determination made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 2 yards (1.8m) full width of the finished cloth. The lot shall be unacceptable if one or more sample units fail to meet any requirements specified. All test reports shall contain the individual values utilized in expressing the final results. The sample size shall be as follows:

Lot size in yards	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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Table III - Test methods

Characteristic	Requirement Paragraph	Test method
Weight	3.3	5041
Yarns per inch (2.5cm)		
Warp	3.3	5050
Filling	3.3	5050
Breaking strength		
Warp	3.3	5100
Filling	3.3	5100
Flame retardancy		
Char length		
Warp	3.5.1	5903
Filling	3.5.1	5903
After flame		
Warp	3.5.1	5903
Filling	3.5.1	5903
After glow		
Warp	3.5.1	5903
Filling	3.5.1	5903
Finish	3.5	4.6.1 1/

1/ A certificate of compliance will be acceptable.

*4.6.1 Compatibility test of Drying Time and Adhesive Strength Specimens of 12 inch (30.5cm) by 12 inch (30.5cm) by 1 inch (2.5cm) sections of fibrous glass insulation board, conforming to type I of MIL-I-742, shall be coated with fire resistant adhesive conforming to class 1 of MIL-A-3316, to an approximate thickness of 1/32 inch (0.8mm). After the coating has dried for 5 minutes, 6 inch (15.2cm) by 12 inch (30.5cm) strips of the cotton brattice cloth shall be applied in such a manner that a strip of cloth covers one-half of the coated section. The cloth shall be pressed and troweled until the adhesive coating extrudes through the interstices and then shall be given a finish coating. The specimens shall then be allowed to dry for 48 hours at approximately 80°F(27° C) room temperature. At the end of this time, three 2 inch (5.1cm) by 6 inch (15.2cm) strips of the covering shall be cut and removed from the surface of the fibrous glass insulation board. The plies of the strips thus obtained shall be stripped back 1 inch (2.5cm) by hand and then pulled to rupture in a tensile testing machine, the jaws of which separate at the rate of 12 inches (30.5cm) per minute. The average results obtained from three strips of specimens shall be reported as the adhesive strength.

*5. PACKAGING

*5.1 Preservation-packaging Packaging shall be level A or C as specified (see 6.2).

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5.1.1 Levels A and C The cloth shall be put up and packaged in accordance with applicable requirements of PPP-P-1135.

5.2 Packing Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C The cloth shall be packed in accordance with the applicable requirements of PPP-P-1135.

5.3 Marking In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1135.

6. NOTES

6.1 Intended use The cloth covered by this specification is intended for repairing and covering the surface of fibrous glass insulation board.

*6.2 Ordering data Procurement documents should specify the following:

a. Title, number and date of this specification.
b. When a first article is required (see 3.1 and 4.3).
c. Width of cloth required (see 3.3.2).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

*6.3 Recycled material It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

*6.4 Changes from previous issue The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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